Sequential Circuits: Latches

In this lab, you learned about the basic building block of sequential circuits: the latch.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Names

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Summary

In this lab, we learned how sequential logic circuits use memory to store previous states. We used Verilog and Vivado to design and test circuits on the Basys 3 board. This helped us better understand how sequential logic works and how inputs affect outputs.

Lab Questions

Why can we not just use structural Verilog to implement latches?

We do not use structural Verilog to implement latches because it is a lot more complicated; doing it the way we did it allows us to decide the latch directly, making it simpler.

What is the meaning of always @(*) in a sensitivity block?

@(*)means the block automatically runs whenever any of its input signals change. It is commonly used for combinational logic in Verilog.

What importance is memory to digital circuits?

Memory allows digital circuits to store information and remember previous states. This makes it possible to count events, perform sequential operations, and control processes.